Analysis of Massive Data Sets

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Finding Frequent Itemsets

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Outline

- **Motivation**
  - Market-Basket model
  - Applications
- **Finding frequent itemsets**
  - Computation model
  - A-Priori algorithm
  - Refinements

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Motivation

Market-Basket Model



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Market-Basket Model

Many-to-many relationship between ***items*** and ***baskets***



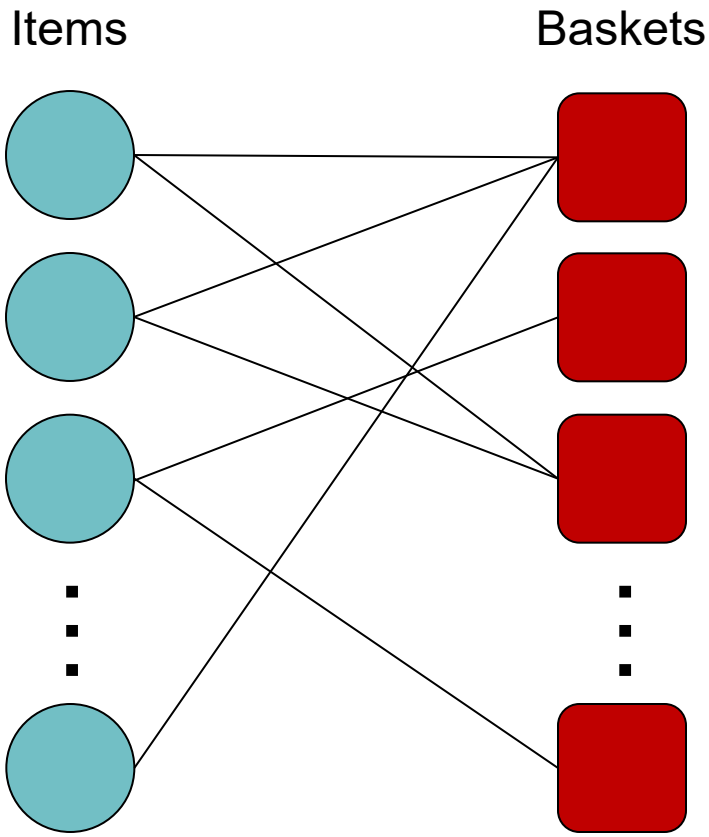
Problem: How to identify items that are ***frequently*** bought together?

Solution: Identification of association rules, i.e. discovery of frequent sets of items (baskets)

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Market-Basket Model

Market-Basket Model: Many-to-many Relationship



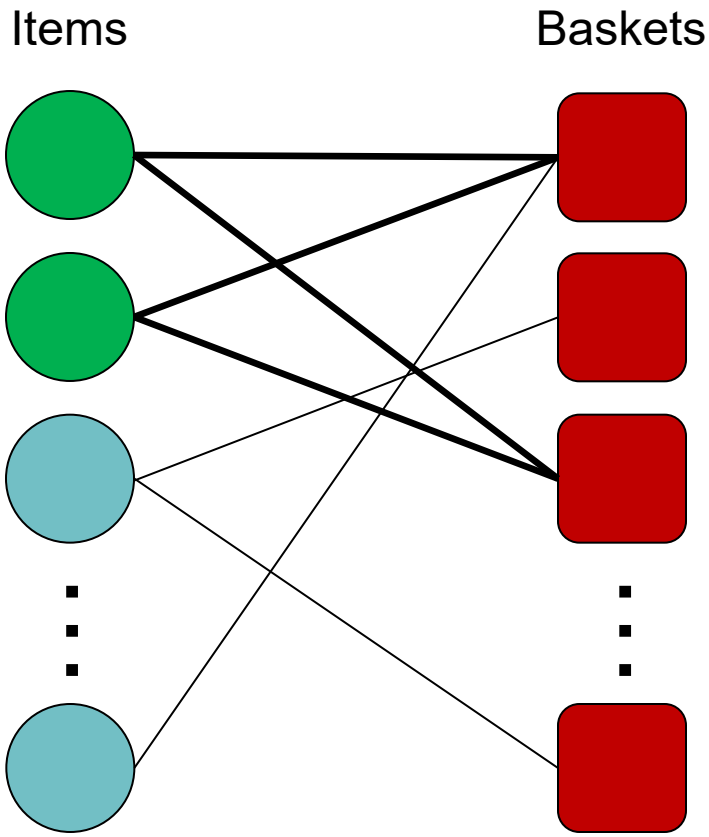
- Large number of items
- Large number of baskets
- Baskets contain a small subset of items

Problem: How many item pairs can be found in at least **k** baskets?

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Market-Basket Model

Market-Basket Model: Many-to-many Relationship



- Large number of items
- Large number of baskets
- Baskets contain a small subset of items

Problem: How many item pairs can be found in at least **k** baskets?

k=2
The two green items are **often** bought together

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Market-Basket Model

The story of beer and diaper

People with small children often buy beer and diapers together as they lack the time to go out

Rule: Put close together beer, diaper + … ? (chips,..)

Trick: Lower the price of beer, rise the price of diapers

Association rules need to be used ***frequently*** in order to be profitable

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Frequent Itemsets vs. Similar Items

Frequent itemsets are about finding the ***absolute*** number a certain itemset appears (***frequency!***)

Finding similar items is about finding very ***similar***, but not necessarily the same sets of items (frequency is **not** important)

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Frequent Itemsets vs. Similar Items

Frequent itemsets – traditional retailers (physical stores)

- Rules need to be frequent in order to be profitable – the store is a physical entity used by all customers

Finding similar items – web stores

- It pays off to know which items could be possibly bought together even by a small number of customers
- Finding correlation between rarely purchased items
- Profiting from the long tail

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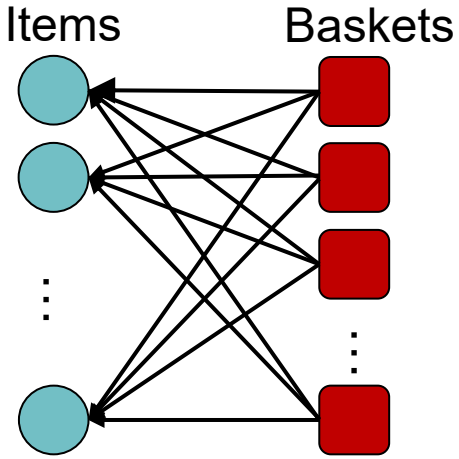
Other Applications

- **Any general many-to-many mapping between two kinds of entities**
- **Detecting drug side-effects**
  - Baskets: patients, Items: drugs and side-effects
- **Related concepts**
  - Baskets: documents, Items: words
    - Smaller documents: tweets, emails
    - Stop words are removed
- **Plagiarism**
  - Baskets: sentences, Items: documents

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Other Applications

- **Biomarkers**
  - Baskets: patient information, Items: biomarkers and diseases
- **Finding Communities in Graphs (e.g. Facebook, Twitter)**
  - Baskets: nodes, Items: outgoing neighbors
  - Searching for **complete** bipartite subgraphs



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Frequent Itemsets

- **Itemset Support**
  - ***Support*** for itemset ***i*** is the number of baskets that contain all items in ***i*** (in the observed dataset)
  - Expressed as an absolute number of baskets or more often as a fraction of the total number of baskets

- **Definition**
  - Given a support threshold ***s***, the item subsets that appear in at least ***s*** baskets are called ***frequent itemsets***

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Frequent Itemsets

- **Items = {beer, chips, juice, wine}**
- **Support threshold = 3 / 6 baskets**

B_1 = {b, c, j, w}

B_2 = {b, c, j}

B_3 = {b, j, w}

B_4 = {c, j, w}

B_5 = {b, c}

B_6 = {b, w}

Frequent itemsets: {b}, {c}, {j}, {w}, {b,c}, {b,j}, {b,w}, {c,j}, {j,w}

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Association Rules

- **Application of frequent itemsets!**
- **If-then rules that describe what a basket is likely to contain**
  - $\{i_1, i_2, i_3, \ldots i_n\} \rightarrow j$ implies that if a basket contains items $i_1, i_2, i_3, \ldots i_n$, it is also likely to contain item j
- **In practice, we want to find rules that really matter**
  - **Support**
  - **Confidence**
  - **Interest**

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Association Rules

- **Association rule support**
  - **support($I \rightarrow j$) = support(I)**
- **Confidence**
  - For $I = \{i_1, i_2, i_3, \ldots i_n\}$
  - ***conf***$(I \rightarrow j)$ **= support**$(I \cup j)$ **/ support**(I)
- **Not all high-confidence rules are interesting**
  - E.g. the rule $X \rightarrow$ *beer* can have high confidence if *beer* is simply purchased very often (independently of X)

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Association Rules

- **Interest**
  - ***Fr* [*j*]** fraction of baskets containing item *j*
  - ***interest*(*I* → *j*) = *conf*(*I* → *j*) - *Fr* [*j*]**

B_1 = {b, c, j, w} B_4 = {c, j, w}

B_2 = {b, c, j} B_5 = {b, c}

B_3 = {b, j, w} B_6 = {b, w}

$c \rightarrow b$

conf (c → b) = 3 / 4

Fr(b) = 5 / 6

interest(c → b) = 3/4 – 5/6

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Association Rules

- Finding association rules
  - Find all association rules with
    - Support ≥ s (support of $i_1, i_2, i_3, \ldots i_n$)
    - Confidence ≥ c
  - If a rule $\{i_1, i_2, i_3, \ldots i_n\} \rightarrow j$ has high support and confidence, both $\{i_1, i_2, \ldots i_n\}$ and $\{i_1, i_2, \ldots i_n, j\}$ will be ***frequent***
    - In realistic scenarios s ≥ 1%, c ≥ 50%
  - Interesting rules
    - Typically, those with interest above 0.5 or below -0.5

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Association Rules

- Finding association rules

1. Find all frequent itemsets ***I*** (taking into account ***s***)

2. Generate rules
   - For every subset ***A*** of ***I*** generate rule $A \rightarrow I \setminus A$
   - Compute **confidence** and select appropriate rules

Key problem: Finding frequent itemsets

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Finding frequent itemsets

- Algorithms
  - A-priori
  - Elcat (Equivalence Class Transformation)
  - FP-growth (Frequent Pattern)

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Finding frequent itemsets

- Computation model
- A-priori algorithm
- Refinements
  - Handling larger data sets in memory
  - Limiting the number of passes

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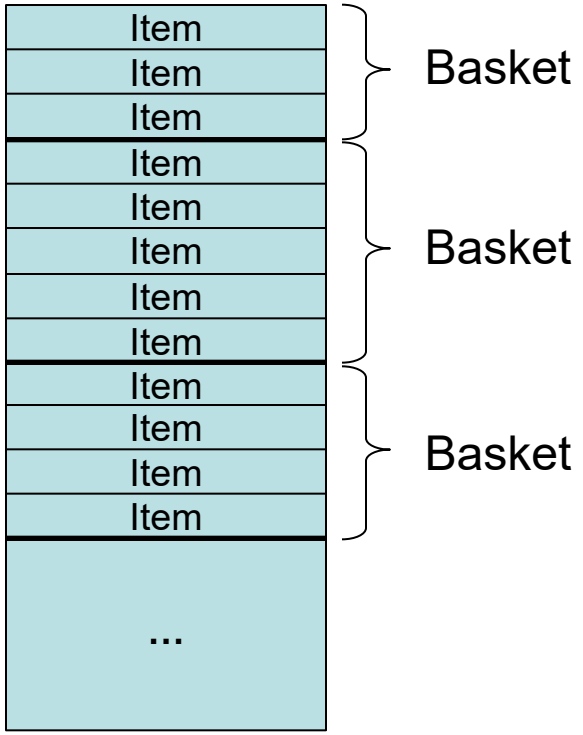
Finding frequent itemsets

- Naïve approach – Frequent pairs
  - Read the data once and count all the pairs
    - For each basket containing **n** items generate all the possible n(n-1) / 2 pairs in two nested loops
  - Issues: although **n** can be small, the overall number of baskets and items can be large
    - 368 million items sold by Amazon.com (Dec 2016)
    - That equals to $67 * 10^{15}$
    - If pairs counts are 4-byte integers, approximately 268 thousand terabytes are needed to store them!

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Computation Model

- **Data Representation**
  - Data is usually stored in flat files
    - Stored on disk (or possibly DFS)
    - Sequences of baskets
    - Average basket size is small



  - The file does not fit into main memory
    - Many items and baskets
    - Baskets are expanded into pairs, triplets, …
    - I/O operations
    - Cost?

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Computation Model

- **Cost of mining frequent itemsets**
  - Counting item subsets in each basket
    - **k** nested loops are used to generate subsets of size **k**
    - Usually, **k** is small and the baskets are small (contain **n** items), e.g. for basket size n = 30 and k = 2 there are $\binom{30}{2} = 435$ pairs
  - I/O operations - often dominant cost
    - Dataset often does not fit into main memory
    - Even if **k** gets large, **n** will decrease as there will be less baskets that meet the support threshold **s**, i.e. the itemsets in baskets are not considered frequent
    - Appropriate support threshold **s** needs to be selected

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Computation Model

- **Main memory bottleneck**
  - Memory needs to be sufficiently large to store frequent itemset counts
  - Algorithms are **limited** by the memory size
    - Swapping the count data would diminish performance by many orders of magnitude

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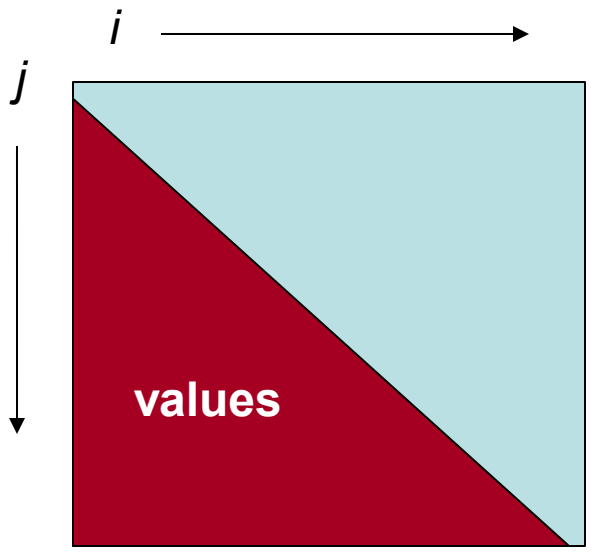
Computation Model

- **How to make it work for massive datasets?**
  - Limit the number of passes an algorithm makes over the data
  - Make frequent itemset counts fit into main memory

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Computation Model

- **Storing counts in memory**
  - Item names should be indexed as integers (hash map)
  - Triangular matrix
    - 2-dimensional array $a[\, i, j \,]$, $i < j$



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Computation Model

- **Storing counts in memory**
  - Triangular matrix
    - More generally one-dimensional array is used
    - Pair { *i*, *j* } is stored at *a*[*k*] for 1 ≤ *i* < *j* ≤ *n* where:

$$k = (i-1)\left(n - \frac{i}{2}\right) + j - i$$

    - If counts are stored as 4 byte integers, approximately $2n^2$ bytes are needed for *n* items

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Computation Model

- **Storing counts in memory**
  - Hash table - Storing triplets
    - Triplet { *i*, *j*, *c* } is stored so that *i* < *j* are item indices and c is the itemset count
    - **Hash table** { *i*, *j* } → *c*
    - If counts are stored as 4 byte integers, 12 bytes are needed to store a singe itemset count
    - However, itemset counts are stored **only** when ***c* > 0**
    - More efficient if significantly fewer then **1/3** of the possible itemsets occur in the baskets (realistic scenario!)

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Computation Model

- **Storing counts in memory**
  - Triangular matrix
  - Storing Triplets (hash table)
  - Still the use of main memory presents limitations as it might not be possible to store the data for all itemsets (pairs, triplets, etc..)
  - Can this situation be improved?

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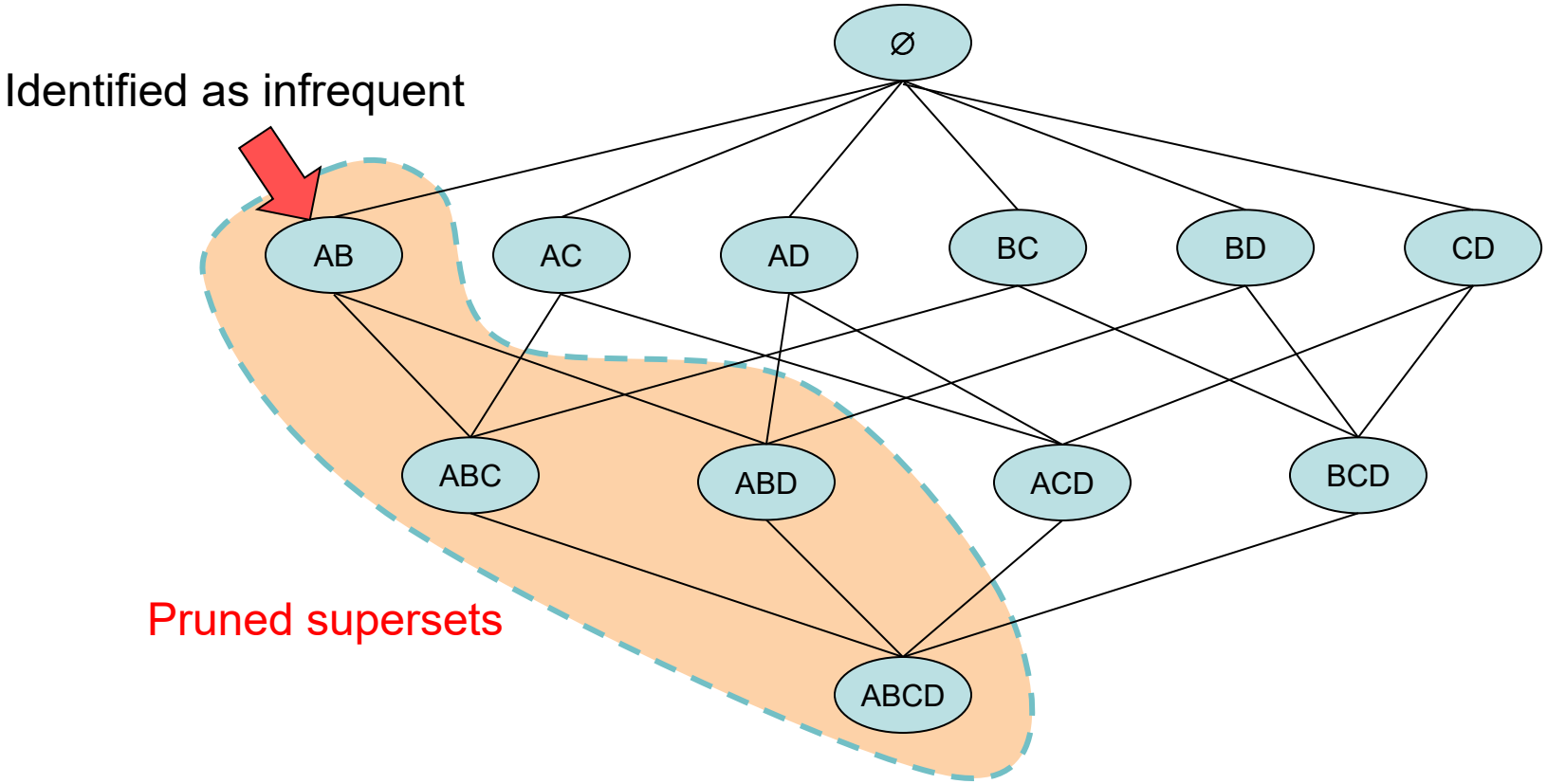
A-Priori Algorithm

- **Rakesh Agrawal (1993)**
- **2-pass algorithm that limits the need for main memory**
- **Key idea: Monotonicity of itemsets**
  - If an itemset ***I*** is **frequent** then each of its **subsets** must also be **frequent**
  - If ***I*** appears ***s*** times then, its subset ***J*** appears at least ***s*** times (it can also appear in other itemsets!)
  - If no superset of an itemset ***I*** is frequent, ***I*** is called the **maximal itemset**
- **Bottom-up approach**
  - start by counting pairs, then triplets, etc.

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A-Priori Algorithm

- **Reducing problem space by pruning infrequent supersets**



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A-Priori Algorithm

- **First pass**
  - Generate a table that translates item names to integers
    - Hash table
  - Count occurrence of each item across all the baskets
    - Array indexed by indices stored in the hash table
    - Size of the array is proportional to the number of items **n**

- **Intermediate step**
  - Generate frequent items table
    - **Scan** through the **counts** and mark the **items** that **meet** the support threshold **s** as **frequent**

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A-Priori Algorithm

- **Second Pass**
  - For each basket look in the frequent items table and see which items are frequent
  - Generate all item pairs using **only** the **frequent items** in the basket
  - Add one to the count for each generated pair in the data structure used to store counts

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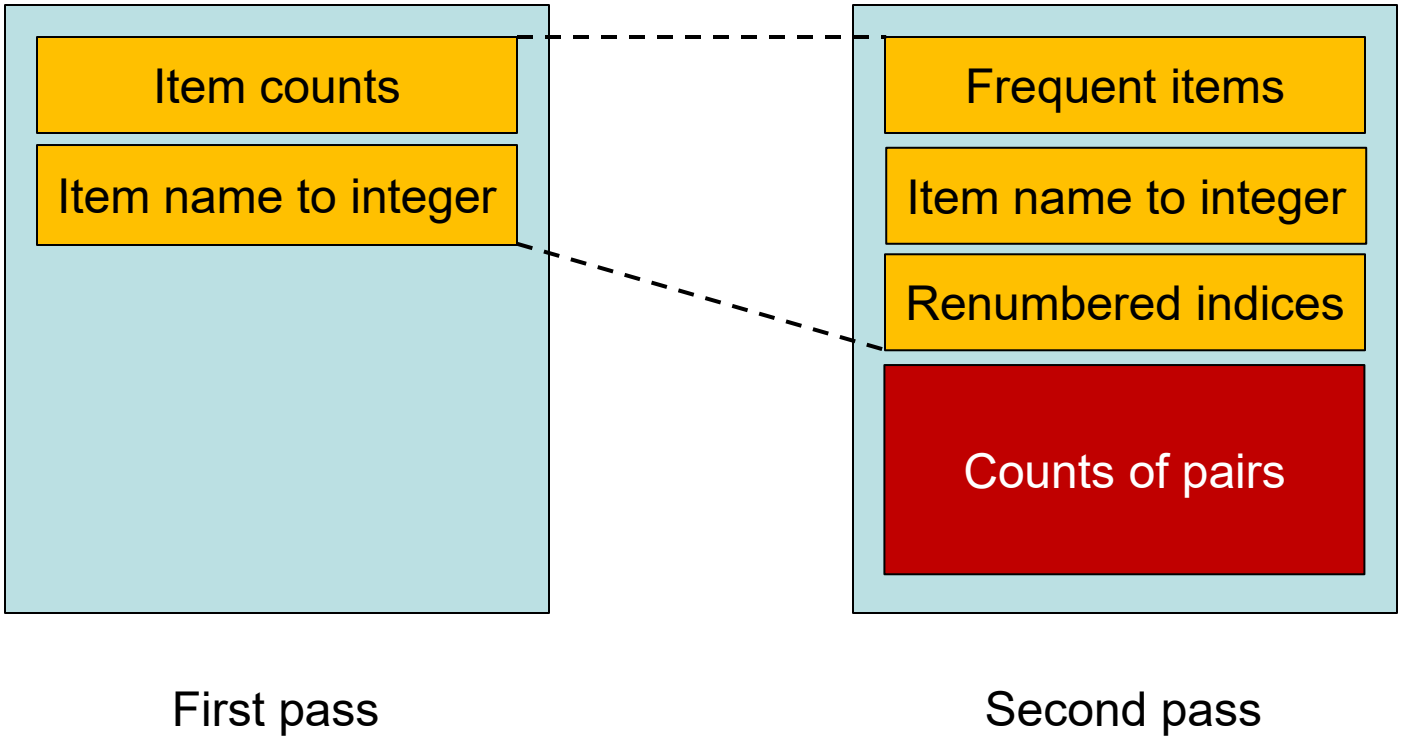
A-Priori Algorithm

- **Second Pass**
  - If the counts are stored in a triangular matrix, items should be renumbered to save space
  - A new table that translates the old indices is added
  - Then the storage size is $2m^2 < 2n^2$ where m is the number of frequent items and n total number of items

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A-Priori Algorithm

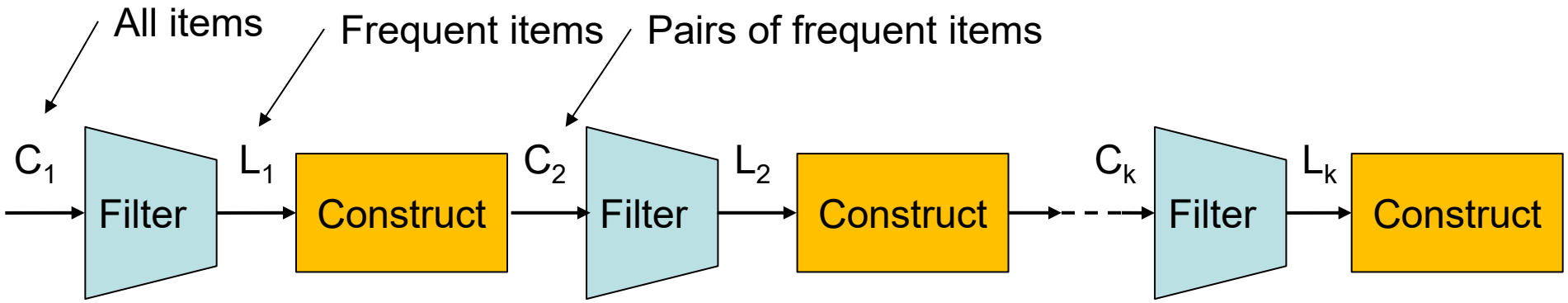
- **Main memory**



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A-Priori Algorithm

- **All frequent itemsets**
  - **k-tuple**: repeat first and second pass ***k*** times
  - $\mathbf{C_k}$ – candidate tuples, have support ≥ s according to step k -1
  - $\mathbf{L_k}$ – filtered C_k so that only frequent k-tuples remain



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A-Priori Algorithm

- **Beyond A-Priori**
  - Limitations on main memory
    - PCY algorithm
  - k passes through the data set are needed to find k-tuples
    - Random sampling
    - SON algorithm
    - Toivonen's algorithm

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PCY Algorithm

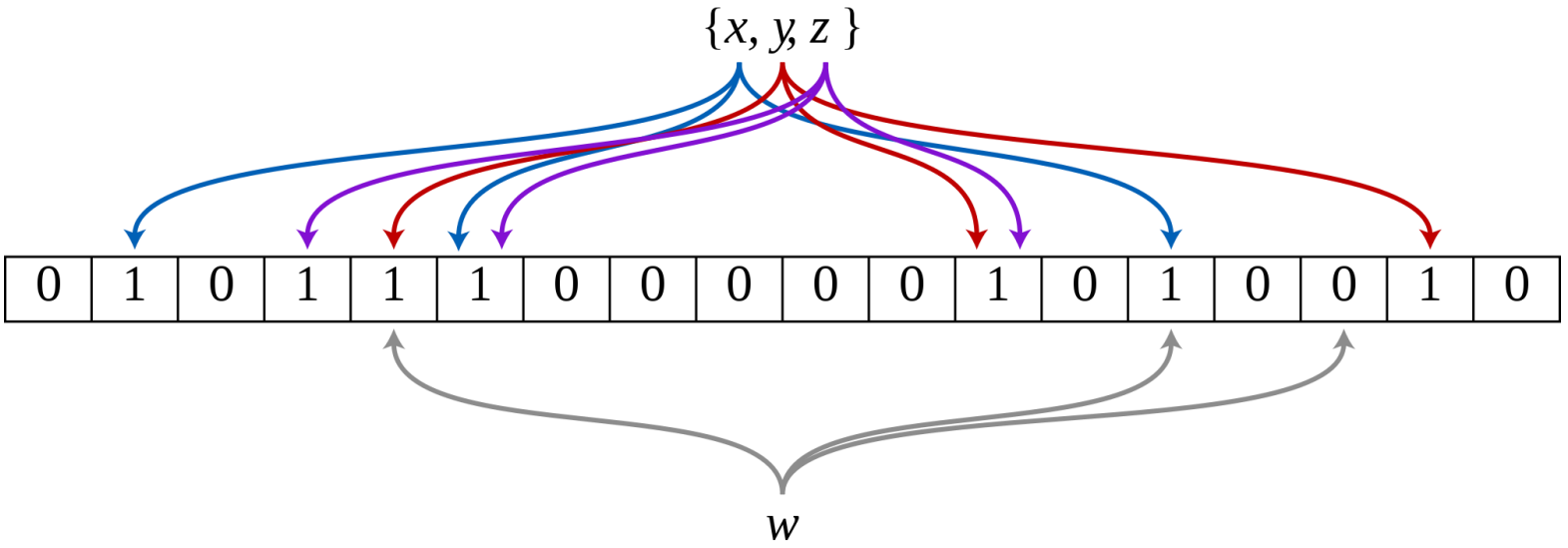
- **Park, Chen, Yu (1995)**
- **Improvement to A-Priori algorithm**

- **Idea**
  - **Most** of the main **memory** in the **first pass** of the A-Priori algorithm is **free**
  - This free space can be used to **reduce the memory requirements** in the second pass (storing itemset counts can be resource demanding)

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PCY Algorithm

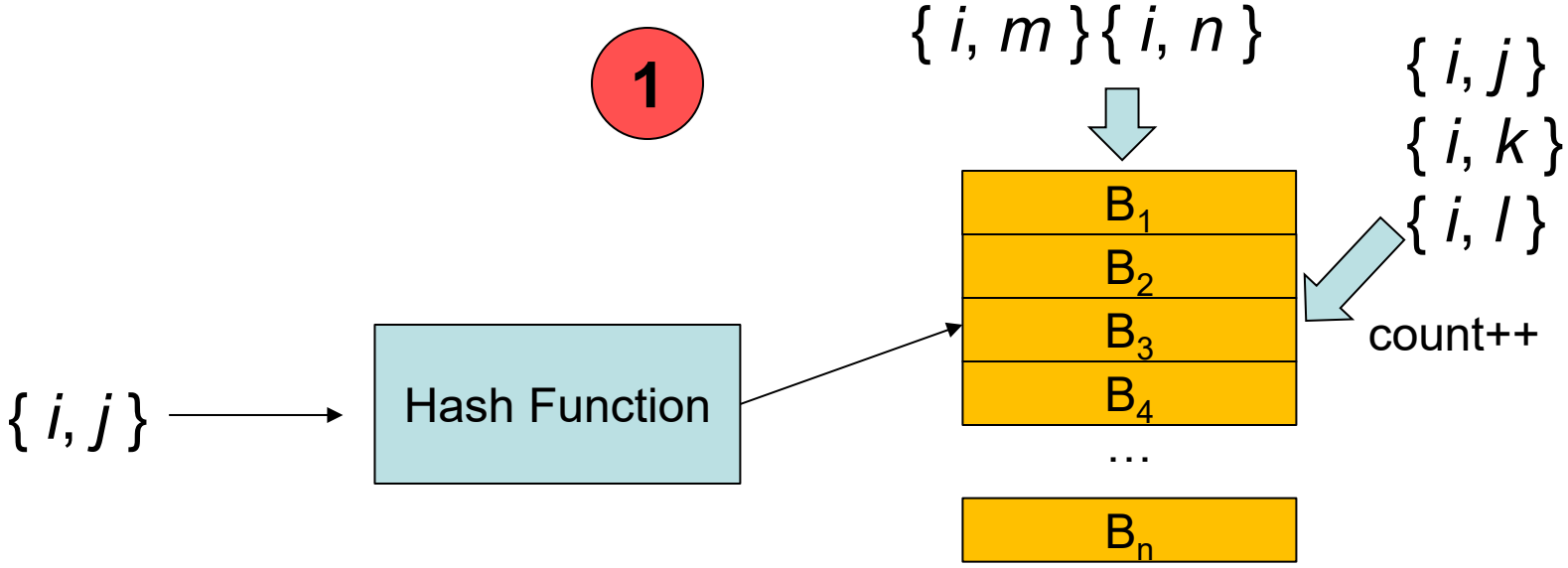
- **How to further reduce memory requirements?**
  - Approach similar to **Bloom filters**
    - Check if an element **is contained in a set**
    - Possible false positives
    - **No false negatives**



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PCY Algorithm

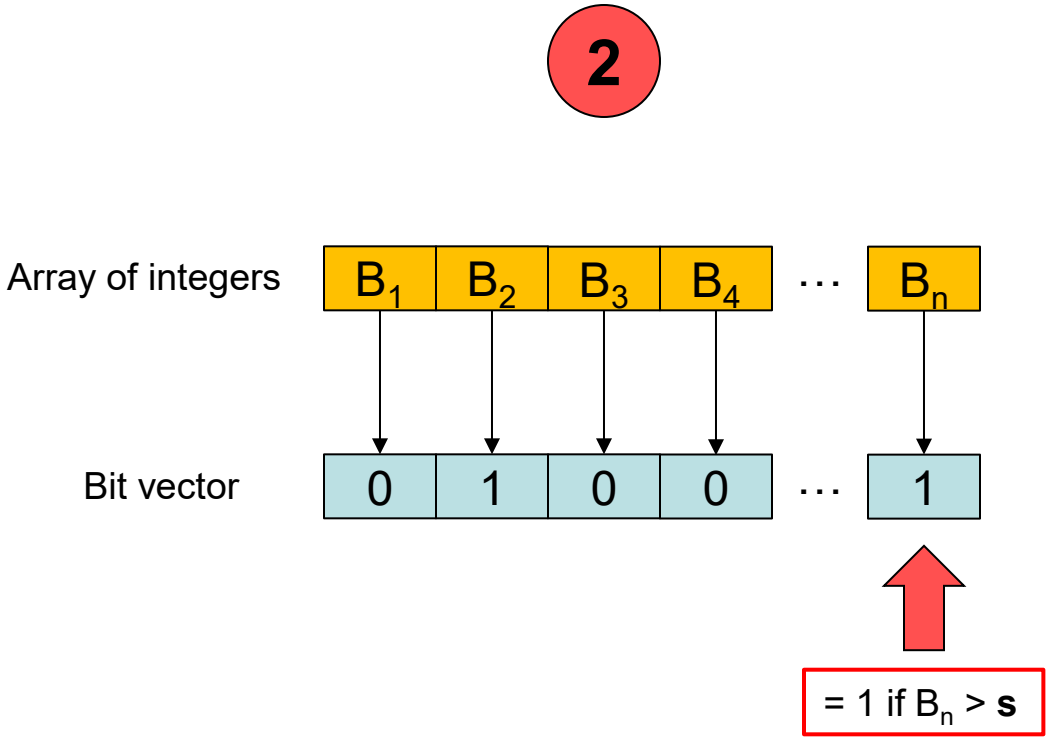
- **How to reduce memory requirements?**
  - Construct a **filter** for infrequent item pairs in the **first pass**



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PCY Algorithm

- **How to reduce memory requirements?**
  - Construct a **filter** for infrequent item pairs in the **first pass**



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PCY Algorithm

- **First pass**
  - Count items, and generate name to index translation table (A-priori)
  - For each **basket** create **item pairs** in a double loop
    - All item pairs in a basket are counted
    - Precision can be increased if only item pairs that consists of frequent items are considered (additional pass needed)
  - **Hash** each item pair into a **bucket** and increase its count
    - There can be as many buckets as the memory permits but crucially there can be less buckets than there are item pairs
    - Bucket counters do not require a lot of space (a few bytes). It is not necessary to count beyond the support threshold.

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PCY Algorithm

- **Intermediate step**
  - Convert bucket counts into a bit-vector
    - Value 1 is assigned to **frequent buckets**, i.e. those that have count greater or equal to support **s**
    - Value 0 is assigned otherwise
  - 4-byte integer counts are replaced by 1bit, therefore a 1/32 of memory is needed to store the bit-vector

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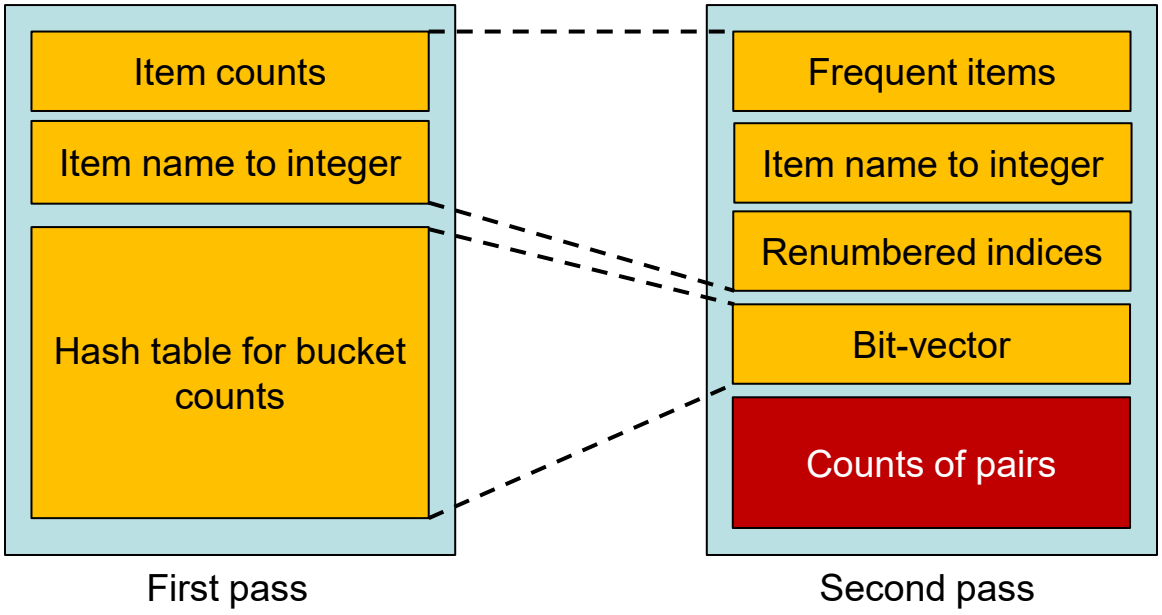
PCY Algorithm

- **Observation on frequent buckets**
  - If a bucket contains a frequent itemset, then it will have count larger or at least equal to **s**, i.e. it will be frequent
    - **However, it can contain 0 frequent itemsets and still be frequent!**
  - If a bucket has count less then **s**, then **none** of the itemsets that hash to it **are frequent**
    - **These itemsets do not need to be counted in the second pass**

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PCY Algorithm

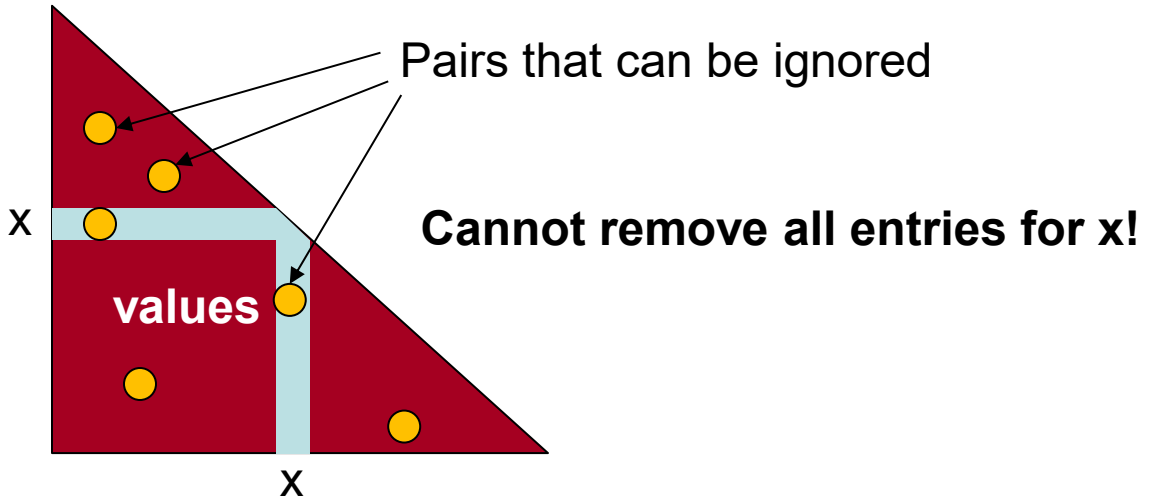
- **Second pass**
  - Count the item pair **{ *i*, *j* }** if and only if:
    - *i* and *j* are **both frequent**
    - The item pair { *i*, *j* } hashes to a bucket that has its bit-vector value set to **1**



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PCY Algorithm

- **Remarks on the PCY algorithm**
  - It usually is more efficient but **not always**
  - Triangular matrix is not suitable to store counts
  - **Why?**
    - Pairs that can be skipped are randomly scattered
    - There is no known way to avoid unused spaces in the matrix



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PCY Algorithm

- **Remarks on the PCY algorithm**
  - Counts should be stored as triplets (e.g. Hash map)
    - PCY is **useful only** if it allows us to avoid counting at least **2/3** of the item pairs
    - Otherwise, A-Priori is more efficient
  - Memory-wise improvements to A-priori, like PCY are generally useful when it comes to finding frequent pairs
  - For triplets, … it is usually sufficient to use A-priori as memory requirements tend to be much lower

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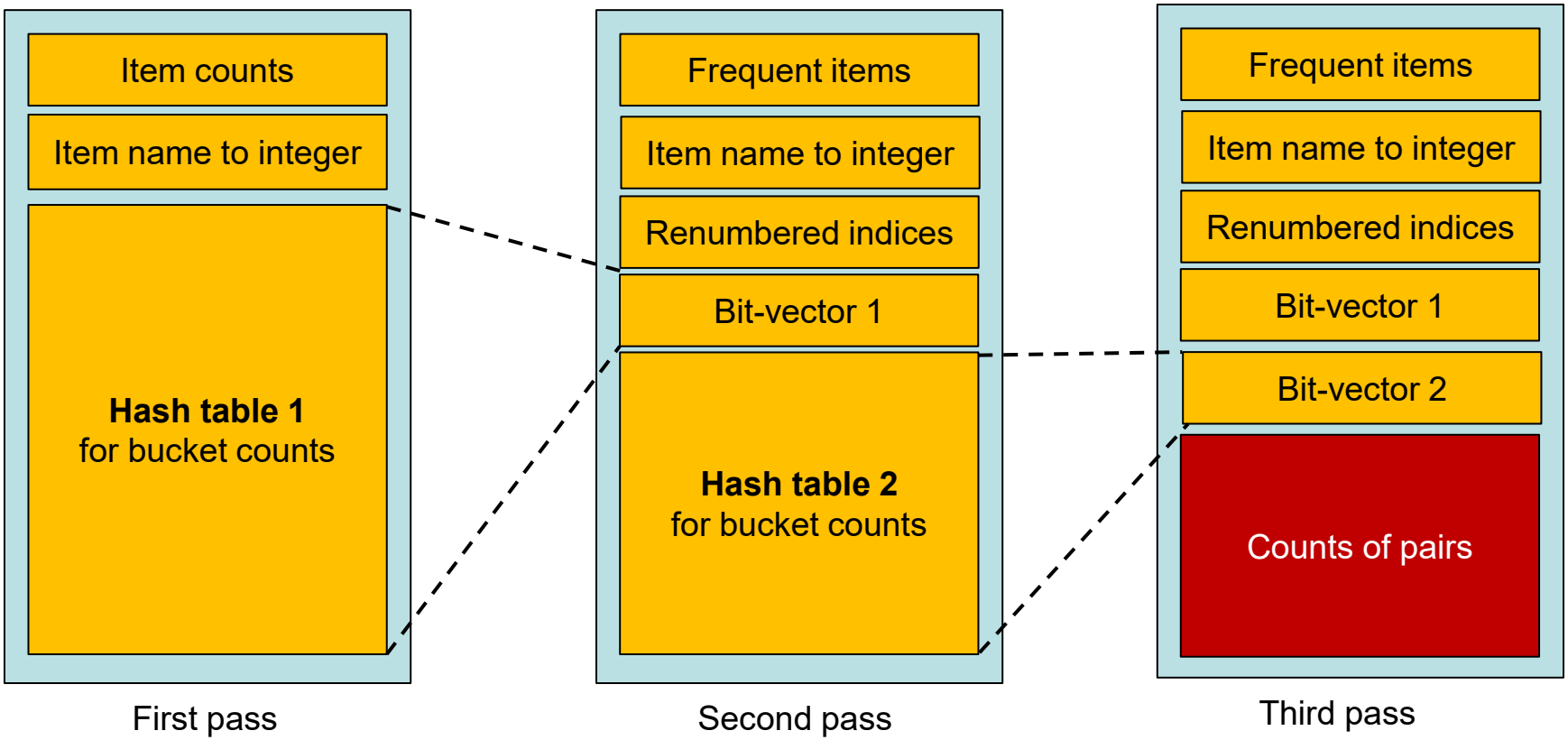
PCY Algorithm

- **Further improvements – Multistage Algorithms**
  - After first pass of PCY item pairs that are candidates to be frequent are **rehashed**
    - New hash function
    - Only pairs that hash to Bit-vector value **1** are counted (further reduces the number of false positives)
  - Number of candidate pairs is reduced, but an **additional pass** through the dataset is **needed**

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PCY Algorithm

- **Further improvements – Multistage Algorithms**



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PCY Algorithm

- **Further improvements – Multistage Algorithms**
  - Pair { *i*, *j* } is counted iff
    - Both i and j are frequent
    - { *i*, *j* } hash to bucket with **Bit-vector 1** value set to 1
    - { *i*, *j* } hash to bucket with **Bit-vector 2** value set to 1

Note: If { *i*, *j* } does not hash to 1 in bit-vector 1 this does not mean that it will not hash to 1 in bit-vector 2 (as other item pairs could have made this particular bucket in bit-vector2 frequent)

- **Thus it is crucial both bit-vectors are checked for each item pair**

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Limiting the Number of Passes

- **Find frequent itemsets in ≤ 2 passes**
  - Fewer passes at the cost of not finding all frequent itemsets
    - This can be acceptable as having too many association rules is not useful
- **Random sampling**
- **SON algorithm**
- **Toivonen's algorithm**

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Random Sampling

- **Randomly select a sample of the dataset**
  - Each entry is selected with probability **p**
    - Sample size is approx. **p** x **n**
  - If baskets are stored randomly
    - First p x n baskets are selected
    - Random DFS chunks are selected

- **Run A-priori or its improvements on the sample in the main memory**
  - Support threshold should be reduced proportionally
    - If sample size is 1% of the dataset s' should be approx. s/100 (the itemset need to appear in s/100 baskets)

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Random Sampling

- **Not all frequent itemsets will be discovered**
  - **False positives**: itemsets frequent in the sample, but not in the dataset
  - **False negatives**: itemsets frequent in the data set, but not in the sample
  - However, if an itemset has support >> **s**, there is a low probability of it being a false negative

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Random Sampling

- **False positives**
  - Can be eliminated by performing an additional pass in which their frequency is verified against **s**
    - Other itemsets do not need to be counted!

- **False negatives**
  - Can be reduced by further decreasing the support threshold and then eliminating false positives
  - Support threshold **s'** = 0.9 **p** x **n**
  - Lowering **s'** increases memory requirements as more items are counted

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Toivonen's Algorithm

- **2 passes**
  - One on the data set sample
  - One on the whole data set
- **No false positives and false negatives!**
- **There is a small probability the algorithm will not produce an answer**
  - It can be repeated and the passes over the data set remain a small constant

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Toivonen's Algorithm

- **Step 1**
  - Generate a random sample
  - If **p** is the portion of the original data set being sampled, **p x s** is the expected support threshold
  - The support threshold is selected to be lower than **p x s**
    - Usually 0.8 – 0.9 p s
    - This increases the memory demands but also reduces the probability that the algorithm will not produce an answer

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Toivonen's Algorithm

- **Step 2**
  - Compute frequent itemsets in the sample
  - Construct the **negative border**
    - Collection of itemsets that are **not frequent** in the sample but **all** their subsets (constructed by removing a single item) **are frequent itemsets**
    - $\varnothing$ is considered to be frequent!

{b, c, j, w}

- Frequent: {b}, {c}, {j}, {b, c}, {b, j}
- Negative border: {w}, {c, j}

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Toivonen's Algorithm

- **Step 3**
  - Pass through entire data set and count:
    - Itemsets frequent in the sample
    - Itemsets in the negative border
  - Outcomes
    1. **No itemsets in the negative border are frequent**
       - Then the itemsets that are confirmed to be frequent on the whole data set are the correct set of frequent itemsets
    2. **At least one itemset in the negative border is frequent**
       - No conclusion can be made – the algorithm needs to restart at Step 1

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Toivonen's Algorithm

- **Why does this approach work?**
  - The main problem are false negatives
  - Assume that **S** is **frequent** in the data set but **not frequen**t in the **sample**
    - This would make it a FN candidate
  - Let **T** be a subset of **S**
    - Smallest subset not frequent in the sample

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Toivonen's Algorithm

- **Why does this approach work?**
  - **T** must be in the negative border
    - If all its immediate subsets are not frequent, then T is not the smallest subset not frequent in sample
  - T is **frequen**t in the **data set**
    - By monotonicity – because S is frequent!
  - Thus, T is frequent and in the negative border
    - Toivonen yields no answer

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SON Algorithm

- **Savasere, Omiecinski and Navathe (1995)**

- **Avoids false positives and negatives**
  - 2 full passes

- **Idea: Divide input into chunks**
  - Chunks are treated as samples
  - Apply random sampling algorithm with support **s'** set to: **p** x **s**, where **p** is chunk size (fraction of the original file)

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SON Algorithm

- **Once chunks are processed**
  - Union of all itemsets that are frequent in at least one chunk
    - These are candidates for frequent itemsets
    - **Any** frequent itemset will **surely be frequent** in at least one chunk – no false negatives
- **Second pass:**
  - Count candidate itemsets and eliminate false positives

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SON Algorithm: MapReduce

- **2 MapReduce passes**

- **First pass**
  - Map
    - **Input:** File chunk, size **p** x **n**, **p** – fraction of input file
    - Count item occurrences and select those with support > **p** x **s**
    - **Output**: **(F_i, 1)** key: F-frequent item, value: irrelevant
  - Reduce
    - **Input:** (F_1, 1), (F_2, 1), (F_3, 1), …, (F_{n-1}, 1), (F_n, 1)
    - **Output:** F_1, F_2, F_3, … , F_{n-1}, F_n

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SON Algorithm: MapReduce

- **Second pass**
  - Map
    - **Input:** portion of input file + (F_1, F_2, F_3, … , F_{n-1}, F_n)
    - Count occurrences of frequent itemset candidates
    - **Output**: **(F_i, v)** key: F-frequent itemset, value: **v** count for **F_i** in the input file chunk
  - Reduce
    - **Input:** (F_1, v_1), (F_2, v_2), (F_3, v_3), …, (F_{n-1}, v_4), (F_n, v_n)
    - Select only those itemsets that have support > **s**
    - **Output:** F_i such that v_i, > **s,** for each **i**

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Literature

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